Web mining End term

Web Mining (Guru Gobind Singh Indraprastha University)



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END TERM EXAMINATION

SEVENTH SEMESTER [B.TECH] DECEMBER 2024

Paper Code: CIE-431T **Subject: Web Mining**

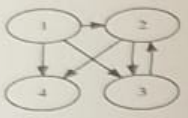
Time: 3 Hours **Maximum Marks: 75**

Note: Attempt all questions as directed. Internal choice is indicated.

Q1 Attempt **any Five** of the following Questions: **(5x5=25)**

a) Describe the concept of association rule mining in web usage mining.

b) What is PageRank? For the following web graph, calculate Page Rank.



c) Describe following social network analysis measures: Centrality and Prestige.

d) Identify and describe three common techniques used in web spamming. How do these methods impact search engine performance?

e) Describe the data pre-processing steps necessary in web usage mining. Why is data cleaning crucial in this context?

f) What is query log mining, and how does it relate to web usage mining?

g) Describe at least three techniques used in web personalization. How do these techniques enhance user experience in e-commerce?

h) Mention any three tools used for webmining. What are their main features and applications in e-commerce and data analysis?

Q2 a) What is a web graph, and how is it utilized in web structure mining? Discuss the significance of hyperlinks in web graphs. **(6.5)**

b) Discuss the various techniques used for extracting and analyzing content from web pages. **(6)**

OR

Q3 a) Define data mining and web mining. How do they differ from each other? Differentiate between web structure mining, web content mining, and web usage mining. **(6.5)**

b) Explain how machine-learning algorithms are applied in data mining. Give examples of popular algorithms used in data mining. **(6)**

Q4 a) What is Latent Semantic Indexing (LSI) and how does it work? What are the limitations of using Latent Semantic Indexing in web content mining? **(6.5)**

b) Define opinion mining and sentiment analysis. Discuss the various techniques used for sentiment classification. **(6)**

OR

Q5 a) Describe the architecture and functioning of a typical web crawler. How do web crawlers deal with duplicate content during the crawling process? **(6.5)**

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b) Define an inverted index. Explain how an inverted index is used to enhance search query performance in web mining. **(6)**

Q6 a) What is classification in web usage mining? What are some common methods for analyzing web usage patterns? **(6.5)**

b) What are the ethical considerations associated with collecting web usage data? **(6)**

OR

Q7 a) Describe the process of building a recommender system. Explain the two main types of collaborative filtering: user-based and item based. What are the strengths and weaknesses of each approach? **(6.5)**

b) Discuss the importance of data cleaning in web usage mining, and list common techniques used for it. **(6)**

Q8 a) Define web data warehousing. Explain the role of OLAP (Online Analytical Processing) in web data warehousing. **(6.5)**

b) Explain the importance of meta-data in web content mining. What challenges are faced when mining large-scale web content? **(6)**

OR

Q9 a) Explain how e-commerce companies can use web mining to enhance customer engagement. What are some techniques for detecting fraud in e-commerce transactions using web mining? **(6.5)**

b) Discuss the role of data integration in e-commerce platforms. Explain the role of ETL (Extract, Transform, and Load) processes in e-commerce data integration. **(6)**

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